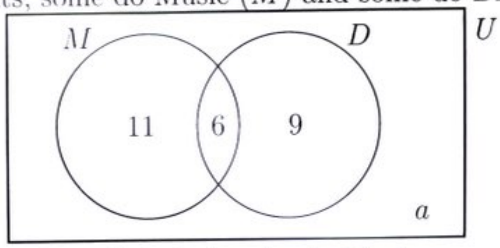
La Scuola d'Italia / Dr. Huson / IB Math: Probability
13 March 2026

First & Last name: CHRIS HUSON

4.19 Quiz: Probability Review

1. In a class of 32 students, some do Music (M) and some do Drama (D).



   (a) Write down the number of students who do Music. [1]

   (b) Find the number of students who do neither Music nor Drama, a. [1]

   (c) One student is chosen at random. Find the probability that

      (i) the student does not do Music; [1]

      (ii) the student does Music or Drama, but not both. [2]

   (d) Given that a student does Drama, find the probability that they also do Music. [2]

a) 17

b) $32-(11+6+9)=6$

c) i) $\frac{9+6}{32}=\frac{15}{32}$

ii) $\frac{11+9}{32}=\frac{20}{32}$

d) $\frac{6}{15}$

2. The following table shows whether 16 students are in the IB math program and whether they have a TI nSpire calculator.

	nSpire	No nSpire	Total
IB	8	2	10
Non-IB	1	5	6
Total	9	7	16

(a) Find the probability that a randomly chosen student is in IB and has a nSpire. [1]

(b) Find P(nSpire | non-IB). [2]

(c) Determine whether "is in IB" and "has a nSpire" are independent events. Show your working and justify your answer. [3]

a) $\frac{8}{16}$

b) $P(nSpire / non\text{-}IB) = \cancel{\frac{8}{7}}\ \cancel{\frac{1}{9}}$

$= 1/6$

c) $P(IB \cap nSpire) = \frac{8}{16}$

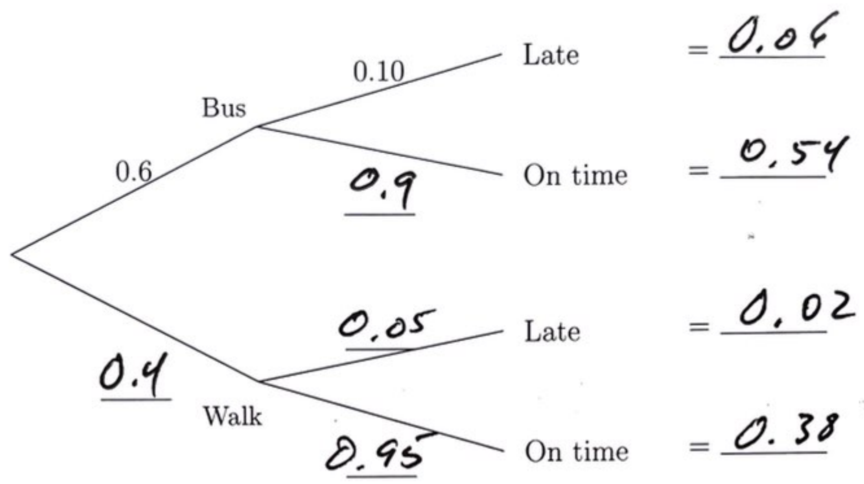
$P(IB) \cdot P(nSpire) = \frac{10}{16} \cdot \frac{9}{16}$

$\frac{10}{16} \cdot \frac{9}{16} \neq \frac{8}{16}$ NOT Independent

First & Last name: Chris Huson

3. At a school, 60% of students take the bus and 40% walk. Of those who take the bus, 10% arrive late. Of those who walk, 5% arrive late.

(a) Complete the tree diagram below, showing all probabilities. [2]



(b) Find the probability that a randomly chosen student walks and arrives late. [1]

(c) Find the probability that a randomly chosen student arrives late. [2]

(d) Given that a student arrives late, find the probability that they walk. [2]

b) $0.04 \cdot 0.05 = 0.02$

c) $0.06 + 0.02 = 0.08$

d) $\frac{0.02}{0.08} = \frac{1}{4}$

4. A bag contains 4 black marbles and 6 white marbles. Two marbles are drawn at random, one after the other, without replacement.

(a) Find the probability that both marbles are black. [2]

(b) Find the probability that both marbles are the same colour. [3]

(c) Find the probability that both marbles are black given that at least one marble is black. [3]

a) $\frac{4}{10} \cdot \frac{3}{9} = \frac{4}{30} = \frac{2}{15}$

b) $P(WW) = \frac{6}{10} \cdot \frac{5}{9} = \frac{1}{3}$

$\frac{4}{30} + \frac{1}{3} = \frac{7}{15}$

c) $P(B \geq 1) = \frac{4}{30} + \frac{4}{10} \cdot \frac{6}{9} + \frac{6}{10} \cdot \frac{4}{9}$

$= 2/3$

$P(BB / B \geq 1) = \frac{4/30}{2/3} = \frac{1}{5}$